

# SHAMITHA YANGALA

Python Backend Developer | 2.9+ Years Experience

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Portfolio: [shamitha-yangala.github.io](https://shamitha-yangala.github.io) | GitHub: [github.com/Shamitha-yangala](https://github.com/Shamitha-yangala) | LinkedIn: [linkedin.com/in/shamitha-yangala](https://linkedin.com/in/shamitha-yangala)

## PROFESSIONAL SUMMARY

Python Backend Developer with 2.9+ years delivering production-grade Django and REST API systems across GIS, healthcare, and HR domains. Reduced manual workloads by up to 75% through workflow automation, cut data estimation effort by 70% via automated proposal generation, and boosted user engagement by 80% on an AI-powered health platform. Strong command of asynchronous task processing, role-based security, containerisation, and database optimisation in high-record-volume environments (10,000+ records).

## TECHNICAL SKILLS

**Languages & Frameworks:** Python | Django | Django | REST Framework | Flask

**Databases:** PostgreSQL | PostGIS | SQLite | Redis

**GIS & Mapping:** GeoDjango | QGIS | LeafletJS | Google Maps API

**DevOps & Cloud:** Docker | Git | AWS (Polly, EC2) | Celery

**Frontend:** JavaScript | AJAX | Bootstrap 5 | HTML5 | CSS3

**AI & Data:** Pinecone | RAG | Pandas | Beautiful Soup | Web Scraping

**Security:** JWT | OAuth2 | Role-Based Access Control | Audit Trails

## PROFESSIONAL EXPERIENCE

### Python Full Stack Developer

SRA OSS India Pvt. Ltd

Feb 2025 – Nov 2025

Hyderabad, India

#### Employee Letter Templates Automation System

- Architected an internal HR platform from scratch, automating generation and tracking of 60+ letter types (Offer, Appointment, Transfer, Revision, Notice) with zero manual document creation.
- Slashed manual HR workload by 75% by replacing paper-based letter workflows with end-to-end digital automation for a team of 200+ employees.
- Engineered conditional Django forms with dynamic field visibility across 60+ letter categories, reducing form-fill errors by an estimated 40%.
- Delivered a salary computation engine that auto-calculates pay structures using 15+ threshold rules and revision data, eliminating payroll calculation errors.
- Designed secure role-based dashboards for Admin, HR, and Staff personas, increasing operational accountability by 30%.

**Tech Stack:** Django, DRF, PostgreSQL, Celery, Redis, Bootstrap 5, JavaScript, AJAX, python-docx, docxcompose

#### Telangana Roads Development Program (Apr 2025 – Aug 2025)

- Delivered GIS dashboard modules enabling real-time road inventory visualisation and data management across 33 districts and 10,000+ road records.
- Cut manual cost-estimation time by 70% by automating 2 major proposal generation workflows using structured survey inputs.
- Accelerated data retrieval speed by 60% through advanced server-side filtering for a dataset spanning 33 districts.
- Streamlined Excel export throughput by 3x using asynchronous background processing with Celery and Redis, supporting bulk exports of 5,000+ rows.
- Configured 4-tier role-based access (Admin, Site Engineer, Inspector, App User) with audit trails and revision logging via Simple History across 10,000+ records.

**Tech Stack:** Python, Django, PostgreSQL, GeoDjango, PostGIS, LeafletJS, Google Maps API, Celery, Redis, Shapely, Pandas, Docker, Git

**Indian Bridge Management System (Feb 2025 – Apr 2025)**

- Customised RESTful GIS APIs to centralise monitoring of bridge inventory, geometry, and inspection data for a national-level infrastructure system.
- Integrated point-based map visualisation serving data for 500+ bridge assets with LeafletJS and djangoestframework-gis.

**Tech Stack:** Django, DRF, QGIS, LeafletJS, Docker, React, djangoestframework-gis

**Python Developer (Remote)**

*Centrix Healthcare Limited*

Feb 2024 – Aug 2024

Mumbai, India

**AI-Powered Cardiovascular Health Application**

- Spearheaded an AI-based health platform using Retrieval-Augmented Generation (RAG) and Pinecone vector database, driving an 80% uplift in user engagement within 3 months of launch.
- Boosted content accessibility for 5,000+ users by integrating AWS Polly text-to-speech across the application.
- Accelerated data acquisition efficiency by 50% by crafting a web scraping pipeline that extracted and structured medical content from 20+ sources.

**Tech Stack:** Django, Flask, PostgreSQL, Pinecone, AWS Polly, BeautifulSoup

**Junior Full Stack Developer**

*E-Systems AI Tech Solutions*

Aug 2022 – Jan 2024

Hyderabad, India

- Contributed to an E-Commerce Platform and Employee Management System serving 1,000+ active users, handling core business logic in Django and Flask.
- Strengthened application security by introducing JWT-based authentication and input validation, reducing reported vulnerability incidents by 35%.
- Optimised critical database queries, achieving a 25% reduction in average API response time across key endpoints.

**Tech Stack:** Python, Flask, Django, HTML, CSS, JavaScript, PostgreSQL

**KEY ACHIEVEMENTS**

- 75% reduction in HR workload — Automated 60+ letter types for a 200+ employee base, replacing all manual document creation.
- 70% faster cost estimation — Automated proposal generation workflows for Telangana Roads Development Program across 33 districts.
- 80% user engagement uplift — Launched an AI-powered RAG cardiovascular health platform at Centrix Healthcare within a 6-month contract.
- 10,000+ records governed — Delivered 4-tier RBAC, audit trails, and revision logging for a national road infrastructure system.

**EDUCATION**

**B.Tech in Computer Science Engineering**

*V.R. Siddhartha Engineering College*

Aug 2017 – May 2020

Vijayawada

**Diploma in Computer Science**

*A.N.M V.V.R.S.R Polytechnic*

Jun 2014 – May 2017

Gudlavalleru